

GPS L1 IF Carrier NCO Generator — Revision 2

Verilog Design & Verification Report — Internship Task 2 (Post-Review)

Revision note: This document supersedes the original Task 2 design report. It reflects two changes requested by the supervisor after reviewing the original submission: (1) `I_out` and `Q_out` changed from 12-bit signed to **3-bit signed**, and (2) the phase accumulator and sine lookup table logic, previously in separate sub-modules, are now **merged into a single `nco_top.v` file** for easier reuse in future tasks.

1. Objective

The task was to design a synthesizable Verilog module implementing a Numerically Controlled Oscillator (NCO) that generates the intermediate-frequency (IF) carrier used in a GPS L1 signal generator/receiver chain. The module produces a 4.092 MHz digital sinusoidal carrier from a 16.368 MHz system clock, using a phase-accumulator-plus-lookup-table architecture, with both I (cosine) and Q (sine) outputs in quadrature (90° apart), now delivered as 3-bit signed samples.

2. Why This Is Needed

A GPS satellite transmits at 1575.42 MHz (L1 band), far too high a frequency to process directly in digital logic. A receiver front-end first down-converts the incoming signal to a much lower intermediate frequency (IF) by mixing it with a locally generated carrier wave, produced digitally by an NCO. Both a cosine (I) and sine (Q) version of the carrier are required because down-conversion is done as a complex (I/Q) multiplication, matching the complex-exponential carrier model described in Section 6.2.2 of the reference text ($\text{carrier} = \exp(j \cdot 2\pi \cdot f_c \cdot t_s \cdot n_n)$).

3. Design Architecture (Revised)

3.1 Single-module structure

The design was originally split into three files: `phase_accumulator.v`, `sine_lut.v`, and `nco_top.v` (which instantiated the first two). Per the supervisor's request, this has been consolidated into a single, self-contained `nco_top.v`: the phase accumulator logic now lives directly inside `nco_top`'s `always` block, and the sine lookup is implemented as a Verilog function within the same module rather than a separate instantiated module. This keeps a single source of truth for the sine table (called twice, once for each of I and Q) while avoiding the overhead of managing multiple files -- intended to make the block easier to drop into future, larger designs as one portable unit.

3.2 Phase accumulator

A free-running 32-bit counter (parameterizable) that adds a fixed value, `phase_step`, to itself on every clock edge and wraps automatically (modulo 2^{32}), representing the carrier's phase angle.

3.3 Sine lookup (function)

A 64-entry table, addressed by the 6 most-significant bits of the phase accumulator, implemented as a Verilog function returning pre-computed values of $\sin(2\pi \cdot \text{addr}/64)$. Values are now scaled to a **3-bit signed**

full-scale of ± 3 (previously ± 2047 for the 12-bit version), per the supervisor's requested output width change. The table is still generated offline (in Python) and embedded as constants in the RTL.

3.4 Quadrature (I/Q) generation

As before, a single sine table (now a function) is evaluated twice per cycle: once addressed directly by the accumulator's MSBs to produce Q (sine), and once addressed with an additional quarter-table offset ($+90^\circ$) to produce I (cosine), exploiting the identity $\cos(x) = \sin(x + 90^\circ)$.

4. Impact of the 3-Bit Output Width Change

A 3-bit signed value has a range of -4 to +3 (8 total levels), far coarser than the previous 12-bit range of -2048 to +2047 (4096 levels). Despite this, the change has no effect on the correctness of this specific design: because the clock-to-carrier ratio (16.368 MHz / 4.092 MHz) is an exact 4:1, only 4 of the 64 table entries (at 0° , 90° , 180° , 270°) are ever visited, and their values are always exactly 0 or the full-scale amplitude. Reducing the output width therefore only rescales the full-scale amplitude from 2047 to 3; the repeating sample pattern shape (full-scale, 0, -full-scale, 0) is unchanged.

Quantity	Previous (Rev 1)	Current (Rev 2)
I_out / Q_out width	12-bit signed	3-bit signed
Representable range	-2048 to +2047	-4 to +3
Full-scale amplitude used	2047	3
Design files	4 (phase_accumulator.v, sine_lut.v, nco_12bit_top.v, nco_12bit_bot.v)	4 (phase_accumulator.v, sine_lut.v, nco_3bit_top.v, nco_3bit_bot.v)
Expected I sequence	2047, 0, -2047, 0, ...	3, 0, -3, 0, ...
Expected Q sequence	0, 2047, 0, -2047, ...	0, 3, 0, -3, ...

5. Frequency Calculations (Unchanged)

The frequency-related calculations are unaffected by this revision:

Quantity	Value / Formula
System clock (Fclk)	16.368 MHz
Target carrier (Fout)	4.092 MHz
Fclk / Fout ratio	$16.368 / 4.092 = 4$ (exact)
phase_step	$2^{32} \times (\text{Fout}/\text{Fclk}) = 2^{32} / 4 = 2^{30} = 1,073,741,824$
LUT address width	6 bits (64 entries)

6. Verification Methodology and Results

The testbench applies synchronous reset for 2 clock cycles, releases it, then captures I and Q on 12 consecutive clock cycles (3 full carrier cycles). Each captured sample is compared against the exact expected value (now using full-scale = 3) for its position in the 4-sample repeating pattern. The self-checking testbench reported **PASS: all 12 samples matched exactly**, and this was confirmed independently in ModelSim, with the waveform showing the expected 3, 0, -3, 0 / 0, 3, 0, -3 staircase pattern for I_out and Q_out respectively.

7. Conclusion

Following supervisor review, the NCO design was revised to (1) output 3-bit signed I/Q samples instead of 12-bit, and (2) consolidate the phase accumulator and sine lookup logic into a single, self-contained nco_top.v module. Both changes were verified to preserve the design's correctness: the exact 4:1 clock-to-carrier ratio means the output pattern shape is unchanged, only its amplitude scale, and the single-file structure produces identical simulation results to the original multi-file version. The revised design passed all 12 self-check samples with zero mismatches.